



HARAMAYA UNIVERSITY
COLLEGE OF COMPUTING AND INFORMATICS

DEPARTMENT OF SOFTWARE ENGINEERING

PROJECT OF REQUIREMENT ENGINEERING
(SRS DOCUMENTS)

TITLE : HU FOTMOB APPLICATIONS

<u>GROUP NAME</u>	<u>IDNO</u>
FIROMSA ABDI	1296/16
ADDISU HIRBO	0576/16
ABUBAKIR ALIYI	0559/16
MOHAMMED HUSSEN	2050/16
DIRO UCHURE	1068/16

DATE : 11/12/2025

SUBMITTED TO : INST. SHUMBEL . A

Table of Contents

1. Introduction	1
1.1 Purpose	1
1.2 Document Conventions	1
1.3 Intended Audience and Reading Suggestions	1
1.4 Product Scope	2
1.5 References	2
2. Overall Description	3
2.1 Product Perspective	3
2.2 Product Functions	4
2.3 User Classes and Characteristics	5
2.4 Operating Environment	6
2.5 Design and Implementation Constraints	6
2.6 User Documentation	7
2.7 Assumptions and Dependencies	7
3. External Interface Requirements	7
3.1 User Interfaces	7
3.2 Hardware Interfaces	8
3.3 Software Interfaces	9
3.4 Communications Interfaces	9
4. System Features	10
4.1 Team & Player Management	10
4.2 Match Scheduling & Fixtures	11
4.3 Real-Time Match Result Entry	11
4.4 Leader boards & Standings	12
4.5 Live Score Display	12
4.6 Tournament History & Records	13
4.7 User Authentication & Access Control	13
4.8 Notifications (Optional)	13
5. Other Nonfunctional Requirements	13
5.1 Performance Requirements	13
5.2 Safety Requirements	14
5.3 Security Requirements	14
5.4 Software Quality Attributes	14
5.5 Business Rules	15
6. Other Requirements	15
6.1 Database Requirements	15
6.2 Legal and Ethical Requirements	16
6.3 Internationalization Requirements	16
6.4 Reporting Requirements	16
6.5 Backup and Recovery Requirements	16
6.6 Reuse Requirements	17
6.7 Future Enhancement Requirements	17
Appendix A – Glossary	17
Appendix B – Analysis Models	18
Appendix C – To Be Determined List	20

1. Introduction

1.1 Purpose

*The purpose of this Software Requirements Specification (SRS) is to describe the requirements for the **HU FotMobApp**, a web-based application designed to efficiently manage and record football tournaments at Haramaya University. This document defines the system's scope, functionalities, constraints, and interactions. It will serve as a reference for developers, project managers, testers, university sports coordinators, and any stakeholders involved in the development and maintenance of the system.*

1.2 Document Conventions

This SRS uses the following conventions:

Bold text is used for headings and key terms.

Italic text emphasizes definitions or important notes.

*Functional requirements are labeled using the format **REQ-X.Y** for clarity and traceability.*

*Priority levels are indicated as **High (H)**, **Medium (M)**, or **Low (L)**.*

Terminology follows definitions provided in the glossary section.

1.3 Intended Audience and Reading Suggestions

This document is intended for the following groups:

Developers: *To understand system requirements and guide implementation.*

Project Managers: *For planning, scheduling, and tracking project progress.*

Testers/Quality Assurance: *To design test cases and validate system behavior.*

Haramaya University Sports Administrators: *To confirm that system functions meet tournament management needs.*

Documentation Writers: *For creating user manuals and help guides.*

Suggested reading order:

Section 1 – Introduction and overview.

Section 2 – Overall description of system components.

Section 3 – Interface and interaction requirements.

Section 4 – Detailed functional features.

Section 5 – Non-functional requirements.

1.4 Product Scope

HU FotMobApp is a modern, web-based system built to streamline football tournament management at Haramaya University. The system automates tasks such as:

Team and player registration

Match scheduling

Real-time score updates

Recording match results

Automatic updating of player statistics (goals, assists, cards)

Displaying leader boards and standings

Generating tournament summaries

Storing historical records for past tournaments

The application improves efficiency by eliminating manual record-keeping and ensuring accuracy, transparency, and accessibility of football-related data. Students and staff can view live updates, team performance, and player histories from anywhere. The system supports the university's initiative to modernize sports management and promote fair, organized competition.

1.5 References

IEEE Software Requirements Specification Template (Karl Wiegars, 1999).

Web application development standards (HTML5, CSS3, JavaScript).

W3C User Interface and Web Accessibility Standards.

Internal project description for HU FotMobApp.

1.6 Definitions, Acronyms, and Abbreviations

A. Definitions

Team : *A group of registered players representing a college, department, or unit within Haramaya University in the football tournament.*

Players : *Individuals who are officially registered under a team. Each player has associated statistics such as goals, assists, and disciplinary records (yellow/red cards).*

Fixture : *A scheduled match between two teams, including date, time, venue, and associated tournament round (group stage, semi-final, final, etc.).*

Leader Board : *A ranked table displaying team or player performance metrics such as points, wins, goals, top scorers, and disciplinary standings.*

Tournament Admin : *An authorized system user responsible for managing tournament operations, including team registration, match scheduling, result entry, and system updates.*

B. Acronyms

Acronym Meaning

SRS	<i>Software Requirements Specification</i>
UI	<i>User Interface</i>
API	<i>Application Programming Interface</i>

C. Abbreviations

Abbreviation	Meaning
HU	<i>Haramaya University</i>
FotMob	<i>Football Mobile (name derived from “Football Mobile Application”)</i>

2. Overall Description

2.1. Product Perspective

HU FotMobApp is a new, web-based tournament management system designed specifically for haramaya University football competitions.

*It replaces traditional manual methods such as paper-based record keeping, **Telegram** announcements, and handwritten match reports.*

*The system acts as a **centralized digital platform** where tournament admins can manage matches, teams, and statistics, while students and staff can access real-time updates.*

The system operates independently but may integrate with:

External data storage (database server)

Notification services (email / push notifications)

Web browsers for client interaction

HU FotMobApp serves as both:

*A **management tool** for administrators*

*An **information platform** for viewers*

2.2 Product Functions

Major system functions include:

2.2.1 Team Management

Register teams and players

Edit team details

2.2.2 Match & Fixture Management

Schedule matches (dates, times, venues)

Update match scores in real time

Manage fixtures automatically or manually

2.2.3 Statistics Tracking

Track goals, assists, yellow/red cards

Update player stats automatically when results are submitted

2.2.4 Live Score Updates

Display ongoing match scores

Refresh results for spectators

2.2.5 Leaderboards & Standings

Team rankings (points, goal difference, wins)

Top player statistics

2.2.6 Historical Records

Store past tournament data

View player and team performance history

2.2.7 User Access & Roles

Admin login for tournament management

Public view for students and staff

2.3 User Classes and Characteristics

2.3.1 Tournament Admins

Manage all tournament activities

Create and update fixtures

Record and edit match results

Moderate team and player data

Skill level: *Moderate computer literacy*

2.3.2 Coaches / Team Managers

Register players

Update team information

Review match schedules and stats

Skill level: *Basic computer literacy*

2.3.3 General Users (Students, Staff, Spectators)

View live scores

Check standings and leader boards

Browse tournament history

Skill level: *Basic browsing ability*

2.3.4 System Administrator (Technical)

Maintains the system

Manages database, bugs, updates

Skill level: *Advanced technical knowledge*

2.4 Operating Environment

The HU FotMobApp will run in the following environment:

Front-End (Client Side):

Any modern web browser: Chrome, Firefox , Edge , Safari

Devices supported: Smartphones , Laptops , Desktops , Tablets

Back-End (Server Side) :

Web server: Apache / Nginx

Database: MySQL / PostgreSQL

Frameworks: PHP, Node.js, or similar web technologies

Network

Requires an internet connection

Accessible over LAN or publicly over the internet

2.5 Design and Implementation Constraints

Web-Based Constraint: *System must run completely in browsers—no installation required.*

Database Constraint: *Must store large historical data securely and efficiently.*

User Role Constraint: *Only authorized admins can modify tournament data.*

Regulatory Constraint: *Must comply with data privacy rules for storing student/player information.*

Technical Constraint: *Supported on common web technologies (HTML, CSS, JS, PHP/Node).
System must be responsive (mobile-friendly).*

Performance Constraint: *Live score updates must appear with minimal delay.*

Resource Constraint: *Running on university servers may impose storage & bandwidth limitations.*

2.6 Assumptions and Dependencies

2.6.1 Assumptions

Admins will provide accurate and timely match results.

Users have basic internet access to use the system.

Tournament structures (group stage, knockout, etc.) follow standard rules.

All team and player data is provided by authorized personnel.

2.6.2 Dependencies

Stable server and database connection

Web browsers used by clients

University internet availability

Third-party libraries/frameworks used in development

University sports department for official data input

3. External Interface Requirements

3.1 User Interface (UI)

The HU FotMobApp provides a clean, responsive, and mobile-friendly web interface that users can access through any modern web browser. The interface should be simple and intuitive for both administrators and general users.

Main UI Components

Home Dashboard : *Displays live scores, fixtures, standings, and latest updates.*

Admin Panel : *Secure section for managing teams, players, fixtures, and match results.
Includes forms for adding/editing data.*

Team & Player Pages : *Shows team details, player lists, and individual player statistics.*

Fixtures Page : *Lists upcoming matches with date, time, and venue.*

Leader boards/Standings : *Displays rankings, points, goal difference, and top scorers.*

Responsive Layout : *UI adapts to phones, tablets, and desktop screens.*

UI Standards

Consistent color theme (HU branding if required).

Standard navigation bar on all screens.

Icons for quick access to features.

Error messages displayed clearly.

Confirmation prompts before deleting or modifying data.

3.2 Hardware Interfaces

Because HU FotMobApp is a web-based system, it requires no special hardware. However, users interact with the system through standard devices:

User Devices

Smartphones (Android, iOS)

Tablets

Laptops

Desktop computers

Server Requirements

Hosting environment capable of running a web server and database (e.g., Apache/Nginx + MySQL/PostgreSQL).

Sufficient storage for historical tournament data.

Stable network connection for real-time updates.

*There are **no direct hardware-to-software physical connections.***

3.3 Software Interfaces

Client-Side Software

Web browsers (latest versions recommended):

Google Chrome

Mozilla Firefox

Microsoft Edge

Safari

Front-end technologies (example):

HTML5, CSS3, JavaScript

Server-Side Software

Web server: APACHE or NGINX

Programming languages: PHP, NODE.JS , or similar

Database: MySQL / PostgreSQL

API layers (if used):

RESTful API for data handling

JSON format for request/response

Third-Party Libraries (optional)

Chart.js or similar for leader boards and stats visualization

jQuery / Axios for asynchronous data loading

Authentication libraries (JWT, OAuth, etc.)

3.4 Communications Interfaces

HU FotMobApp uses web-based communication protocols for data exchange between the client and server.

Protocols

HTTP/HTTPS for secure communication

TCP/IP as underlying network protocol

Real-Time Communication

If real-time score updates are implemented:

Web Sockets or periodic AJAX calls may be used.

Data Formats

JSON for sending and receiving data

HTTPS for encrypted data transmission

Network Requirements

Internet or local network access

Stable bandwidth for live updates

4. System Features

4.1 Team & Player Management

4.1.1 Description and Priority : *This feature allows tournament admins to register new teams, add players, update details, and manage rosters.*

Priority: High

4.1.2 Stimulus/Response Sequences : *Admin selects “Add Team” → System shows team registration form → Admin submits → System saves team.*

Admin selects “Add Player” → Enters player info → System validates and saves.

Admin edits team/player info → System updates database.

4.1.3 Functional Requirements

REQ-1: *The system must allow admins to create, edit, and delete teams.*

REQ-2: *The system must allow admins to add players under a specific team.*

REQ-3: *The system must validate required fields before saving team/player data.*

REQ-4: *The system must store player attributes (name, number, position, department).*

REQ-5: *Only authorized admins can modify team and player information.*

4.2 Match Scheduling & Fixtures

4.2.1 Description and Priority : *Allows admins to generate and manage match fixtures with date, time, venue, and participating teams.*

Priority: High

4.2.2 Stimulus/Response Sequences : *Admin clicks “Create Fixture” → Selects teams → Sets date/time → System creates the match schedule.*

Admin updates schedule → System adjusts listing.

4.2.3 Functional Requirements

REQ-6: *The system must allow admins to create fixtures with required match details.*

REQ-7: *The system must prevent scheduling conflicts (same team, same time).*

REQ-8: *The system must display fixtures to all users in chronological order.*

REQ-9: *The system must allow admins to edit or cancel a scheduled match.*

4.3 Real-Time Match Result Entry

4.3.1 Description and Priority: *Admins can record live match results including goals, assists, and cards.*

Priority: High

4.3.2 Stimulus/Response Sequences: *Admin enters ongoing match data (goals, cards) → System updates score in real time.*

Score appears on the live scoreboard for all viewers.

4.3.3 Functional Requirements

REQ-10: *The system must allow admins to input goals, assists, and disciplinary actions.*

REQ-11: *The system must update the match score instantly.*

REQ-12: *The system must lock results once the admin marks the match as “Completed.”*

REQ-13: *The system must store match events for historical retrieval*

4.4 Leaderboards & Standings

4.4.1 Description and Priority: *Displays team rankings and player statistics based on recorded match data.*

Priority: High

4.4.2 Stimulus/Response Sequences : *System receives updated match data → Automatically recalculates ranking → Displays updated leader board.*

4.4.3 Functional Requirements

REQ-14: *The system must automatically calculate team points, goal difference, wins, and draws.*

REQ-15: *The system must display top scorers and top performers.*

REQ-16: *The system must refresh the leader board after every match update.*

REQ-17: *The leader board must be visible to all users.*

4.5 Live Score Display

4.5.1 Description and Priority : *Shows real-time scores for ongoing matches to all viewers.*

Priority: Medium–High

4.5.2 Stimulus/Response Sequences : *Admin updates match → Viewers instantly see changes on the live score page.*

4.5.3 Functional Requirements

REQ-18: *The system must display live scores for ongoing matches.*

REQ-19: *The live score page must refresh automatically or use real-time updates (Web Socket/AJAX).*

REQ-20: *Users should be able to view match events (goal, card, substitution)*

4.6 Tournament History & Records

4.6.1 Description and Priority : *Stores all past tournaments, match results, team performance, and player histories.*

Priority: Medium

4.6.2 Stimulus/Response Sequences : *User selects “Past Tournaments” → System displays historical data. User selects a specific match → System shows details and statistics.*

4.6.3 Functional Requirements

REQ-21: *The system must store all completed tournament data.*

REQ-22: *Users must be able to view past match results.*

REQ-23: *The system must allow search/filtering by year, team, or tournament type.*

4.7 User Authentication & Access Control

4.7.1 Description and Priority : *Controls access to admin features and protects sensitive data.*

Priority: High

4.7.2 Stimulus/Response Sequences : *Admin logs in → System verifies credentials → Grants access to admin dashboard.*

5. Other Non-Functional Requirements

5.1 Performance Requirements

PERF-1: *The system must load main pages (home, fixtures, leader board) within 3 seconds under normal network conditions.*

PERF-2: *Live score updates should appear within 1–2 seconds after the admin enters a new match event.*

PERF-3: *The system must support at least 200+ concurrent users during peak tournament hours.*

PERF-4: *The database should handle large historical records (multiple seasons) without performance degradation.*

PERF-5: *Queries for standings, fixtures, and team data must return results within 2 seconds.*

5.2 Safety Requirements

SAFE-1: *The system must ensure that stored tournament data (teams, players, results) is not lost due to server failure by implementing regular backups.*

SAFE-2: *Only authorized users should be able to modify sensitive data (match results, player stats).*

SAFE-3: *The system must prevent accidental deletion of critical records by providing confirmation prompts.*

SAFE-4: *In case of unexpected shutdown, the system must not corrupt ongoing match records.*

5.3 Security Requirements

SEC-1: Admins must authenticate using a secure login system before accessing management features.

SEC-2: All passwords must be stored using encrypted or hashed methods.

SEC-3: The system must use **HTTPS** to protect data transmission between client and server.

SEC-4: Access control must ensure that only admins can edit team, player, or match data.

SEC-5: The system must log suspicious activities such as repeated failed login attempts.

SEC-6: Sensitive user information must not be visible to unauthorized users.

5.4 Software Quality Attributes

Usability

QUAL-1: The UI must be simple and intuitive for users with basic computer skills.

QUAL-2: The interface must be responsive and usable on mobile, tablet, and desktop devices.

Reliability

QUAL-3: The system must be available at least **95%** of the time during active tournament seasons.

QUAL-4: Data must remain consistent even when multiple admins are updating information simultaneously.

Maintainability

QUAL-5: Code must be modular to allow easy updates and bug fixes.

QUAL-6: Documentation should be available for future maintenance.

Portability

QUAL-7: The system must run on all major browsers (Chrome, Firefox, Edge, Safari).

QUAL-8: The system should be deployable on different web servers without major changes.

Scalability

QUAL-9: *The system must be able to expand to support additional tournaments, more teams, and more users in the future.*

5.5 Business Rules

BR-1: *Only authorized tournament admins can register teams and update match results.*

BR-2: *A match cannot be marked as “Completed” unless scores and key events are fully recorded.*

BR-3: *Leaderboards must follow official tournament point rules (e.g., Win = 3, Draw = 1, Loss = 0).*

BR-4: *Team or player data must remain accurate and cannot be altered without admin authentication.*

BR-5: *Each team must have a unique name within a tournament season.*

6. Other Requirements

6.1 Database Requirements

DB-1: *The system must maintain a structured database storing teams, players, matches, results, and statistics.*

DB-2: *The database must support relational integrity (foreign key constraints between teams, players, and matches).*

DB-3: *Historical data must be preserved and never overwritten when new tournaments begin.*

DB-4: *The system must support database backups and restoration procedures.*

DB-5: *Sensitive data (admin accounts, player details) must be stored securely.*

6.2 Legal and Ethical Requirements

LEGAL-1: *The system must comply with basic data privacy rules when handling student information.*

LEGAL-2: *User data should not be shared with third parties without authorization.*

LEGAL-3: *The system must avoid storing unnecessary personal information about players or users.*

LEGAL-4: *Copyright must be respected when using team logos, images, or university assets.*

6.3 Internationalization Requirements (If Needed)

INT-1: *The system must support English as the default language.*

INT-2: *The system should be adaptable to add more languages in the future if required.*

(Optional feature — depending on university plans)

6.4 Reporting Requirements

REP-1: *The system should generate printable reports for match results, standings, and player statistics.*

REP-2: *Reports must be exportable in formats such as PDF or CSV (optional).*

REP-3: *Admins should be able to generate full-season summaries.*

6.5 Backup and Recovery Requirements

BCK-1: *The system must perform automated database backups daily or weekly (depending on admin preference).*

BCK-2: *Backup files must be stored securely on the server or cloud storage.*

BCK-3: *The system must allow database restoration in case of server failure.*

6.6 Reuse Requirements

REU-1: *The system design should follow modular coding practices to allow reuse of components such as login module, leader board logic, and match update API.*

REU-2: *Reusable UI components (cards, tables, forms) should be applied throughout the system.*

6.7 Future Enhancement Requirements

(These are not part of the initial release but may be added later.)

FUT-1: *Mobile app version for Android/iOS.*

FUT-2: *Automated referee assignment and attendance tracking.*

FUT-3: *Integration with HU student ID system for player verification.*

FUT-4: *Live commentary or event timeline during matches.*

FUT-5: *Online fan voting for "Player of the Match."*

Appendix A: Glossary

This glossary defines terms, acronyms, and abbreviations used throughout the SRS.

Terms

Team – *A group of players representing a department or unit in the HU tournament.*

Player – *An individual registered to participate in the tournament, with associated statistics.*

Fixture – *A scheduled match including date, time, and venue.*

Leader board – *A ranking table showing team points, standings, and player statistics.*

Tournament Admin – *Authorized personnel responsible for managing tournament data and system operations.*

Live Score – *Real-time update of match events such as goals and cards.*

Match Event – *Any recorded action during a match (goal, assist, yellow/red card).*

Acronyms

SRS – *Software Requirements Specification*

UI – *User Interface*

API – *Application Programming Interface*

Abbreviations

HU – *Haramaya University*

FotMob – *Football Mobile (system brand name).*

Appendix B: Analysis Models

*This section includes high-level models used during system analysis.
(These do not need to be fully detailed diagrams unless required by your instructor.)*

1. Context Diagram (High-Level View)

HU FotMobApp sits at the center of interactions between:

Tournament Admin (inputs match data, fixtures, teams)

General Users (view scores, stats, standings)

Database Server (stores all records)

2. Use Case Model (Summary)

Key use cases:

Admin logs in

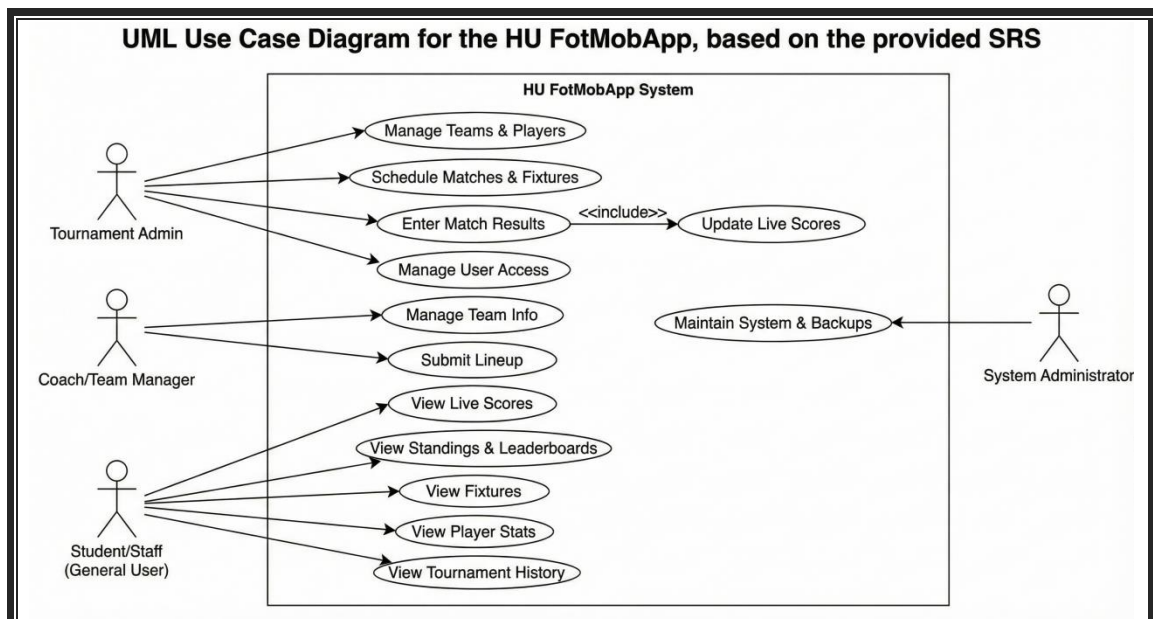
Admin registers teams and players

Admin creates fixtures

Admin updates live scores

System updates leaderboards

Users view results, standings, and player stats



3. Entity Relationship Model (Simple Textual Form)

Entities:

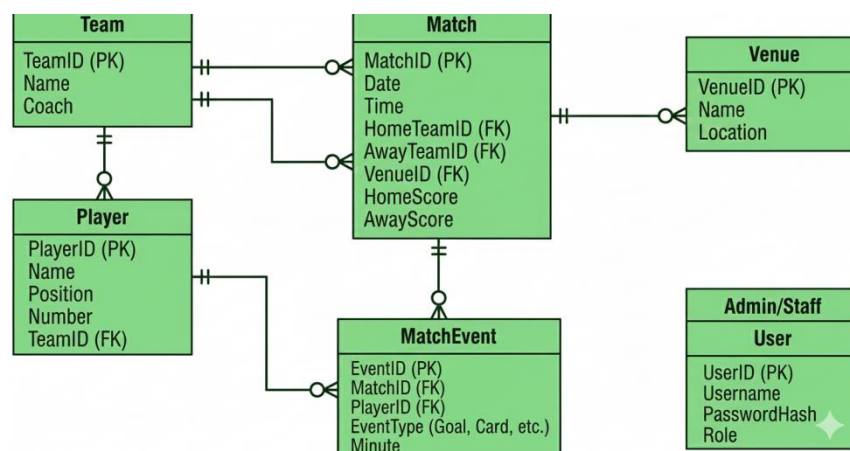
Team (*team_id, name, coach, department*)

Player (*player_id, team_id, name, position, number*)

Match (*match_id, teamA, teamB, date, venue, status*)

Match Event (*event_id, match_id, player_id, type, time*)

Leaderboard (*team_id, points, wins, draws, losses, goal_diff*)



4. Data Flow Overview

Admin enters data → System updates database → UI displays updated standings and scores.

User requests information → System retrieves data → UI displays results.

(If you want actual diagrams (ERD, DFD, Use Case Diagram), tell me and I will generate them in clean text or image format.)

Appendix C: To Be Determined (TBD) List

This section lists items not yet defined but required for completion in future iterations.

TBD No.	Description	Notes
TBD-1	Exact UI color theme and branding	Pending HU approval
TBD-2	Final hosting server (cloud, on-premise, or campus server)	Decision required by ICT office
TBD-3	Notification system (email, SMS, push)	Optional future enhancement
TBD-4	Match venue list and official scheduling rules	Awaiting Sports Directorate input
TBD-5	Data backup frequency (daily/weekly)	To be set by system admin